

COMP-CO835

Object Oriented Systems

Week 10 Lecture

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Outline

- The System Development Life Cycle
- Methodologies, Models, Tools, and Techniques
- Agile Development
- The Unified Process, Extreme Programming, and Scrum
- Principles of Project Management
- Project Management Body of Knowledge (PMBOK)

Learning Objectives

- Compare the underlying assumptions and uses of a predictive and an adaptive system development life cycle (SDLC)
- Explain what makes up a system development methodology—the SDLC as well as models, tools, and techniques
- Describe the key features of Agile development
- Understand and describe the key features of the Unified Process, Extreme Programming, and Scrum Agile system development methodologies

Learning Objectives

- Describe the factors that cause a software development project to succeed or fail
- Describe the responsibilities of a project manager
- Describe the knowledge areas in the project management body of knowledge (PMBOK)
- Explain the Three level approach (three rings) of Extreme Programming (XP) in systems development projects

Overview

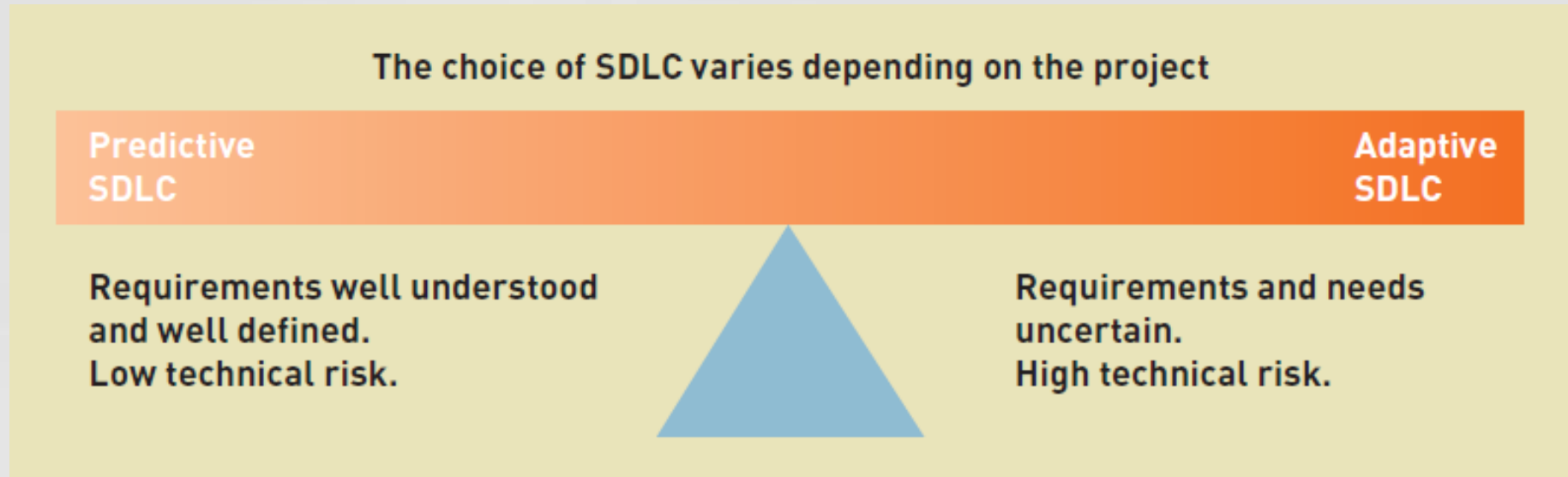
- Review of the SDLC and related concepts in more detail
 - Predictive versus Adaptive SDLC variations
 - Activities and Tasks of System Support
 - Models, Methodologies, Tools and Techniques
 - Agile Development
- Specific SDLC Approaches
 - Unified Process (UP)
 - Extreme Programming (XP)
 - Scrum

The System Development Life Cycle (SDLC)

- There are two general approaches to the SDLC
- Predictive Approach to the SDLC
 - Waterfall model
 - Assumes the project can be planned in advance and that the information system can be developed according to the plan
 - Requirements are well understood and/or low technical risk
- Adaptive Approach to the SDLC
 - Iterative model (as seen in this text)
 - Assumes the project must be more flexible and adapt to changing needs as the project progresses
 - Requirements and needs are uncertain and/or high technical risk

The System Development Life Cycle (SDLC)

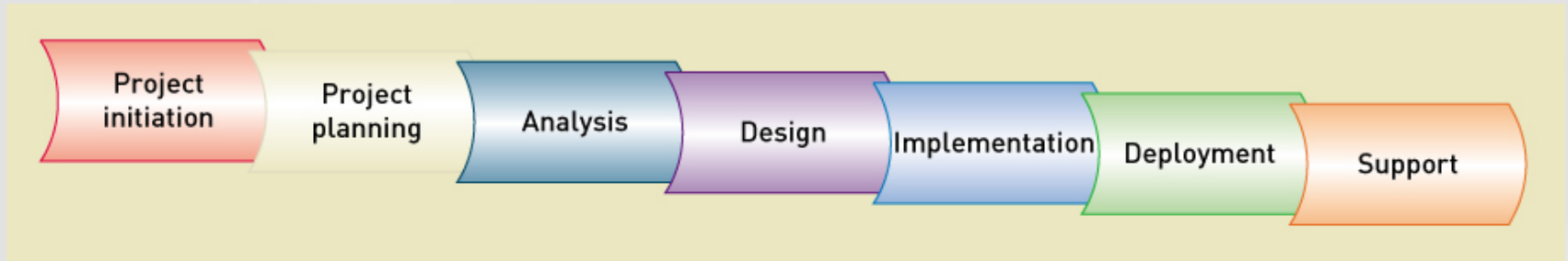
- Most projects fall on a continuum between Predictive and Adaptive



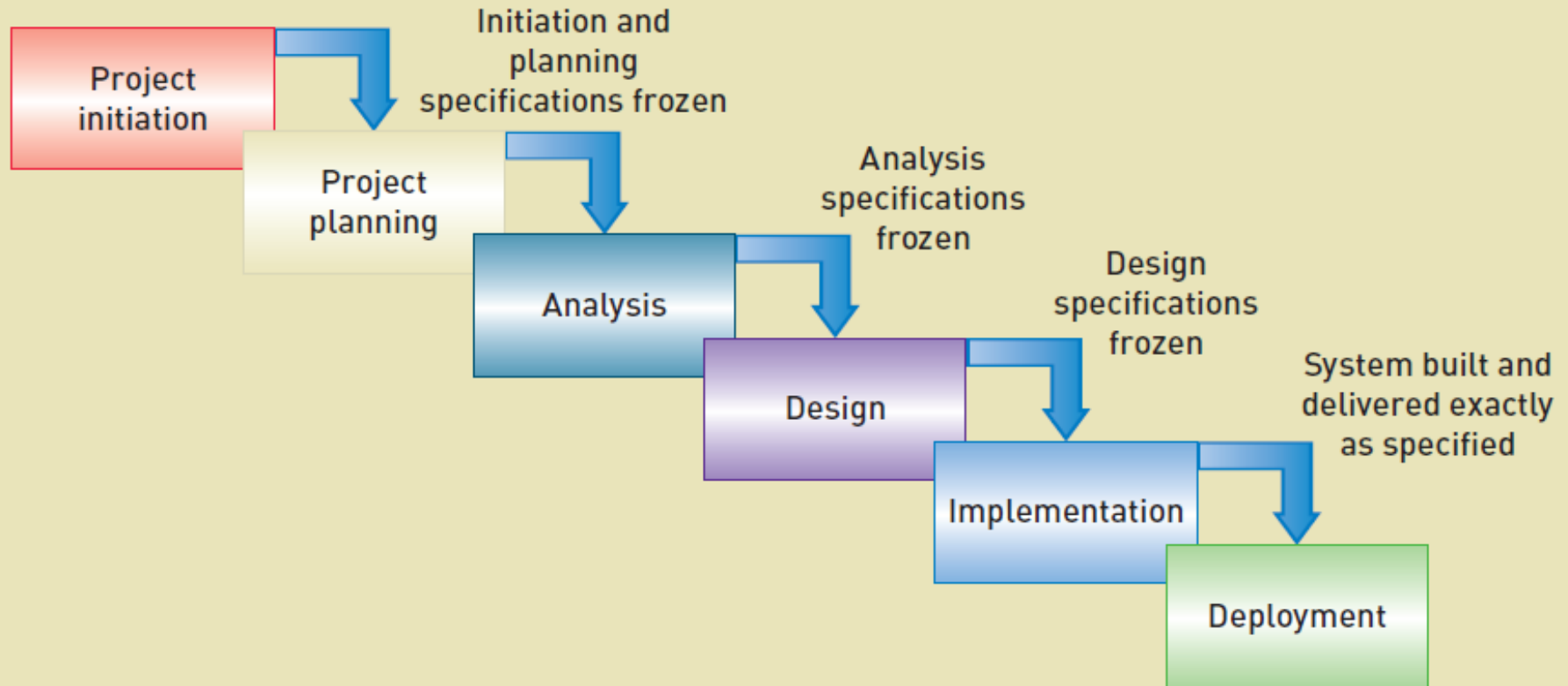
Traditional Predictive SDLC

- Earlier approach based on engineering
- Typically have sequential *Phases*
 - Phases are related groups of development activities, such as planning, analysis, design, implementation, and deployment
- Waterfall model
 - SDLC that assumes phases can be completed sequentially with no overlap or iteration
 - Once one phase is completed, you fall over the waterfall to the next phase, no going back

Tradition IS Development Phases



Waterfall Predictive SDLC



Adaptive Approaches

- Incremental development
 - Completes portions of the system in small increments and integrated as the project progresses
 - Sometimes considered “growing” a system
- Walking Skeleton
 - The complete system structure is built first, but with bare-bones functionality

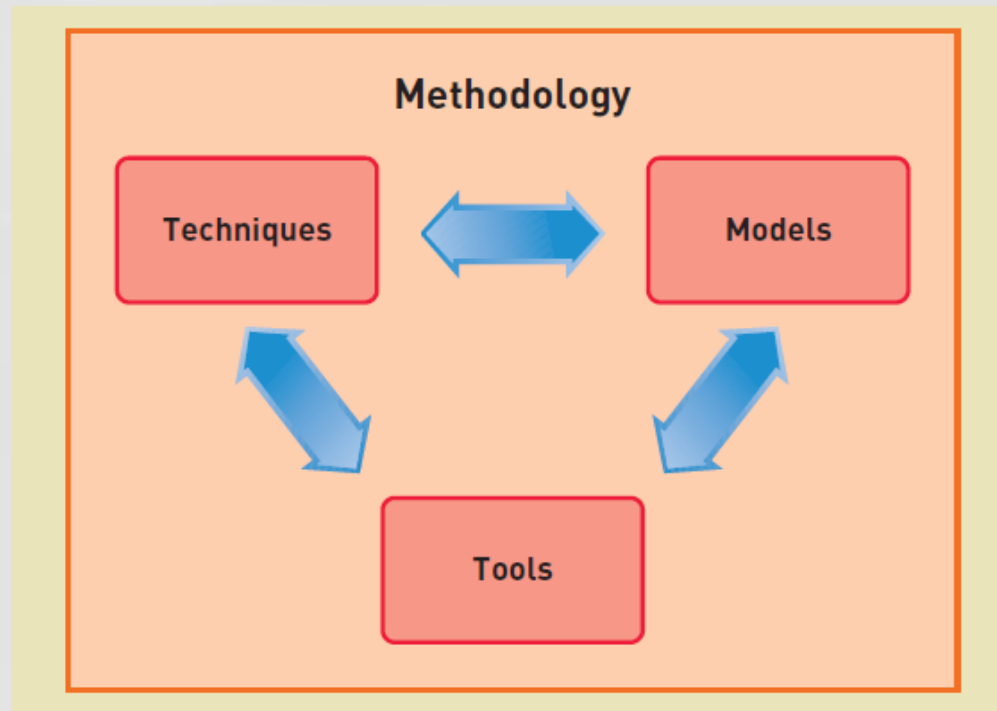
Methodologies, Models, Tools, and Techniques

● Methodologies

- Provides guidelines for every facet of system development: What to do when, why and how
- Specifies the SDLC with activities and tasks
- Specifies project planning and project management methods and reporting
- Specifies analysis and design models to create
- Specifies implementation and testing techniques
- Specifies deployment and support techniques

Methodologies

- A **Methodology** includes a collection of techniques that are used to complete activities and tasks, including modelling, for every aspect of the project



Methodologies, Models, Tools, and Techniques

- Model
 - An abstraction of an important aspect of the real world.
 - Makes it possible to understand a complex concept by focusing only on a relevant part
 - Each model shows a different aspect of the concept
 - Crucial for communicating project information
- In IS, some models are of system components that will be developed
- Other models are used to manage the development process

Methodologies, Models, Tools, and Techniques

Some models of system components

- Use case diagram
- Domain model class diagram
- Design class diagram
- Sequence diagram
- Package diagram
- Screen design template
- Dialog design storyboard
- Entity-relationship diagram (ERD)
- Database schema

Some models used to manage the development process

- Gantt chart
- Organizational hierarchy chart
- Financial analysis models—NPV, payback period
- System development life-cycle model
- Stakeholders list
- Iteration plan

Methodologies, Models, Tools, and Techniques

• Tools

- Software applications that assist developers in creating models or other components required for a project

• Integrated Development Environment (IDE)

- set of tools that work together to provide a comprehensive development environment

• Visual Modelling Tools

- Tools to create graphical models

Methodologies, Models, Tools, and Techniques

● Some typical tools

Project management application
Drawing/graphics application
Word processor/text editor
Visual modeling tool
Integrated development environment (IDE)
Database management application
Reverse-engineering tool
Code generator tool

Methodologies, Models, Tools, and Techniques

● Technique

- A collection of guidelines that help an analyst complete an activity or task
- Learning techniques is the key to having expertise in a field

Strategic planning techniques
Project management techniques
User-interviewing techniques
Data-modeling techniques
Relational database design techniques
Structured programming techniques
Software-testing techniques
Process modeling techniques
Domain modeling techniques
Use case modeling techniques
Object-oriented programming techniques
Architectural design techniques
User-interface design techniques

Agile Development

- A guiding philosophy and set of guidelines for developing information systems in an unknown, rapidly changing environment

Agile Development

- Agile Values in “Manifesto for Agile Software Development”
 - Value responding to change over following a plan
 - Value individuals and interactions over processes and tools
 - Value working software over comprehensive documentation
 - Value customer collaboration over contract negotiation

Agile Modeling Principles

- Develop software as the primary goal
 - Don't get distracted by documentation or models
- Enable the next effort as your secondary goal
 - Be aware of next step versions or revisions
- Minimize your modeling activity
 - Only build what helps move the project forward
- Embrace change and change incrementally
 - Take small steps that keep you on-track and that can be reversed if necessary

Agile Modeling Principles

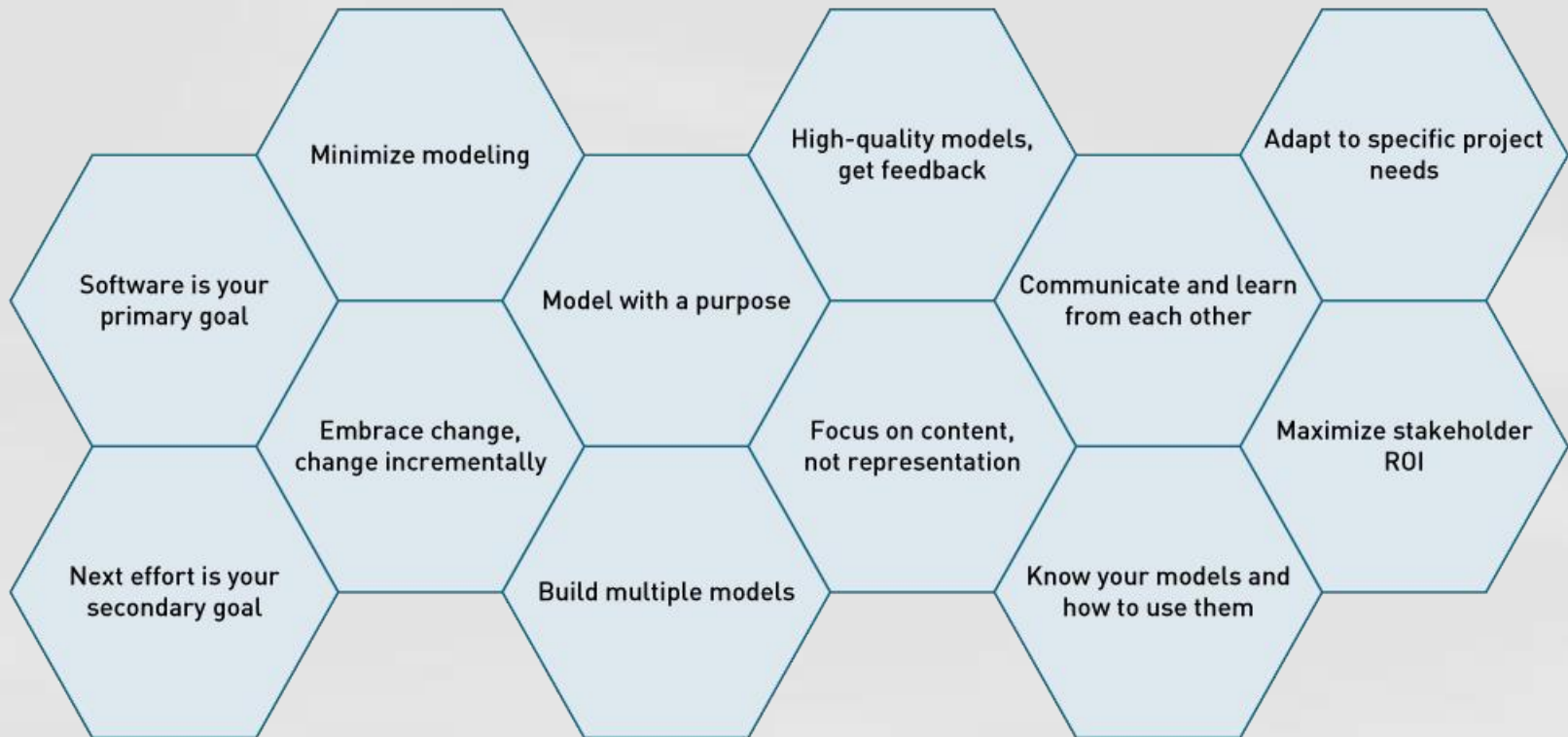
- Model with a purpose
 - Model to understand
 - Model to communicate
- Build multiple models
 - Look at problems from different perspectives
- Build high-quality models and get feedback
- Focus on content rather than representation
 - Informal hand-drawn models are sometimes okay
 - Always focus on stakeholder needs

Agile Principles

- Learn from each other with open communication
- Know your models and how to use them
- Adapt to specific project needs
- Maximize stakeholder ROI

Agile Modeling Principles

- Agile Modeling (AM) – 12 Principles
 - A philosophy – build only necessary models that are useful and at the right level of detail



The Unified Process (UP)

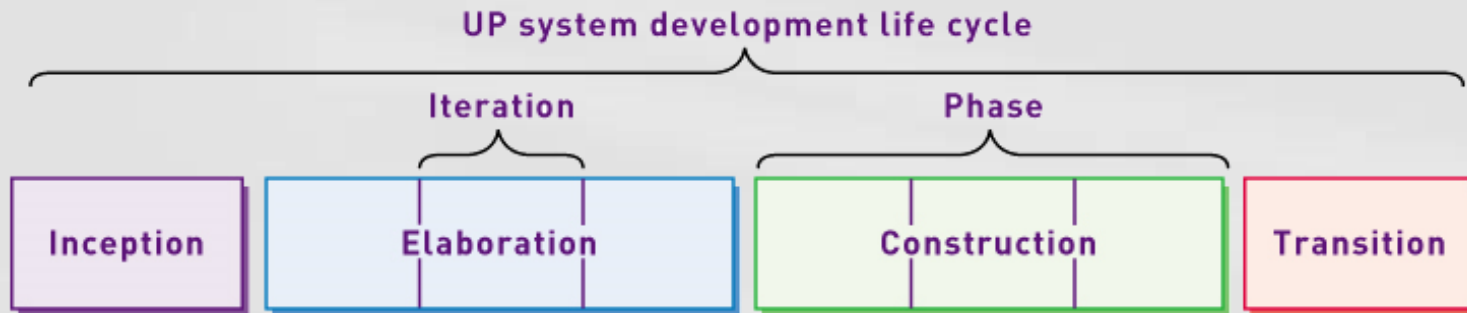
- The UP was the early leader in adaptive approaches
- UP and UML (Unified Modeling Language) were developed together

The Unified Process (UP) – Phases

- UP Phases organize iterations into four primary areas of focus during a project
 - Inception phase – getting the project started
 - Elaboration – understanding the system requirements
 - Construction – building the system
 - Transitions – preparing for and moving to deploying the new system

Unified Process (UP) – Phases

- UP SDLC divides iterations into four phases



Unified Process – Disciplines

- A set of functionally related development activities
 - Each discipline are all the activities related to achieving one objective in the development project
 - Two types of disciplines
 - Development disciplines
 - Management – planning and control disciplines

Unified Process – Disciplines

- Development Disciplines
 - Business modeling
 - Requirements
 - Design
 - Implementation
 - Testing
 - Deployment

Unified Process – Disciplines

- Planning and Control disciplines
 - Configuration and change management
 - Project management
 - Environment

Extreme Programming (XP)

- Takes the best practices of software development and extends them “to the extreme”
 - Focus intensely on proven industry practices
 - Combine them in unique ways to get better results
- XP Core Values
 - Communication
 - Simplicity
 - Feedback
 - Courage

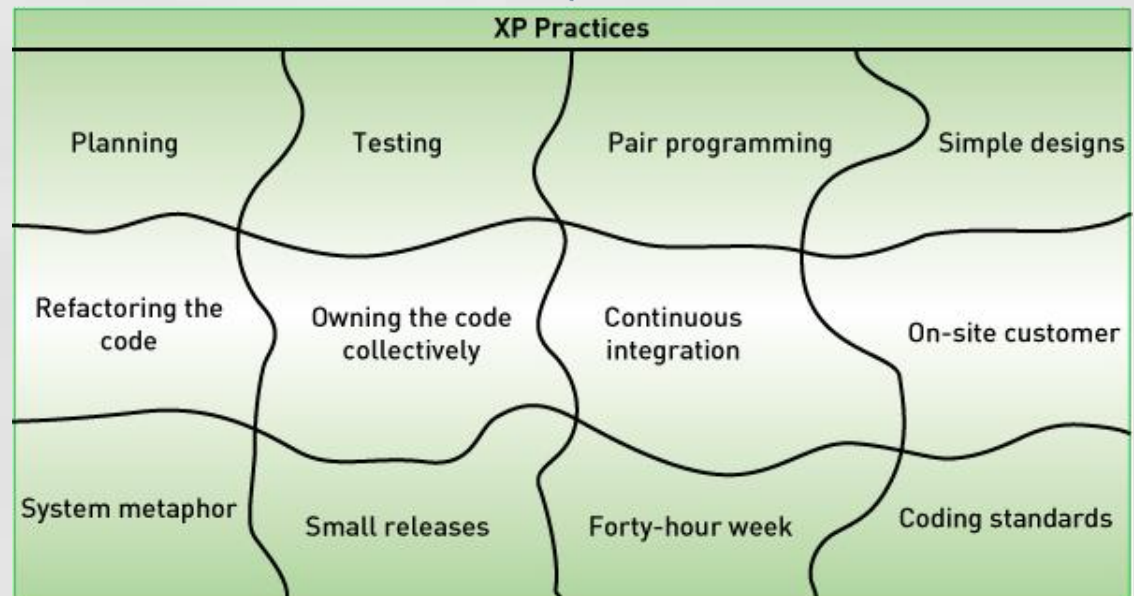
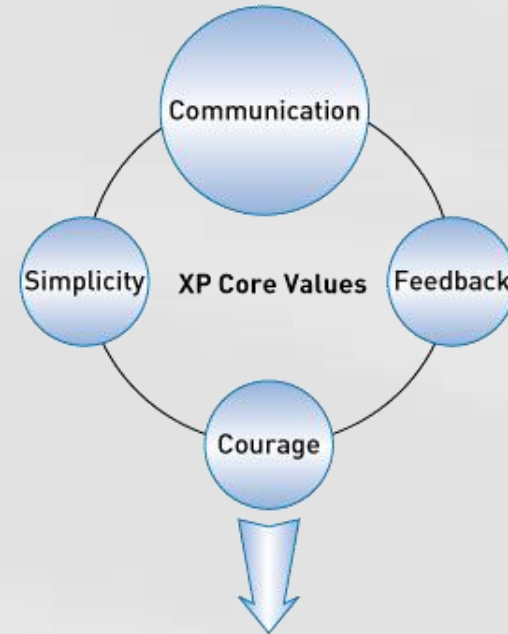
Extreme Programming (XP)

- XP Practices (1 of 2)
 - Planning – based on user stories
 - Testing – thorough testing at every step
 - Pair Programming – watch, inspect, trade off
 - Simple Designs – Agile modeling principles
 - Refactoring – redo and cleanup as you go
 - Owning the code collectively – egoless development, anyone can review and improve code

Extreme Programming (XP)

- XP Practices (2 of 2)
 - Continuous integration – grow the software continuously
 - On-site customer – get sign-off as you go
 - System metaphor – what should the final system look like
 - Small releases – turn over to user frequently
 - Forty-hour work week – don't overload the developers
 - Coding standards – follow standards for code

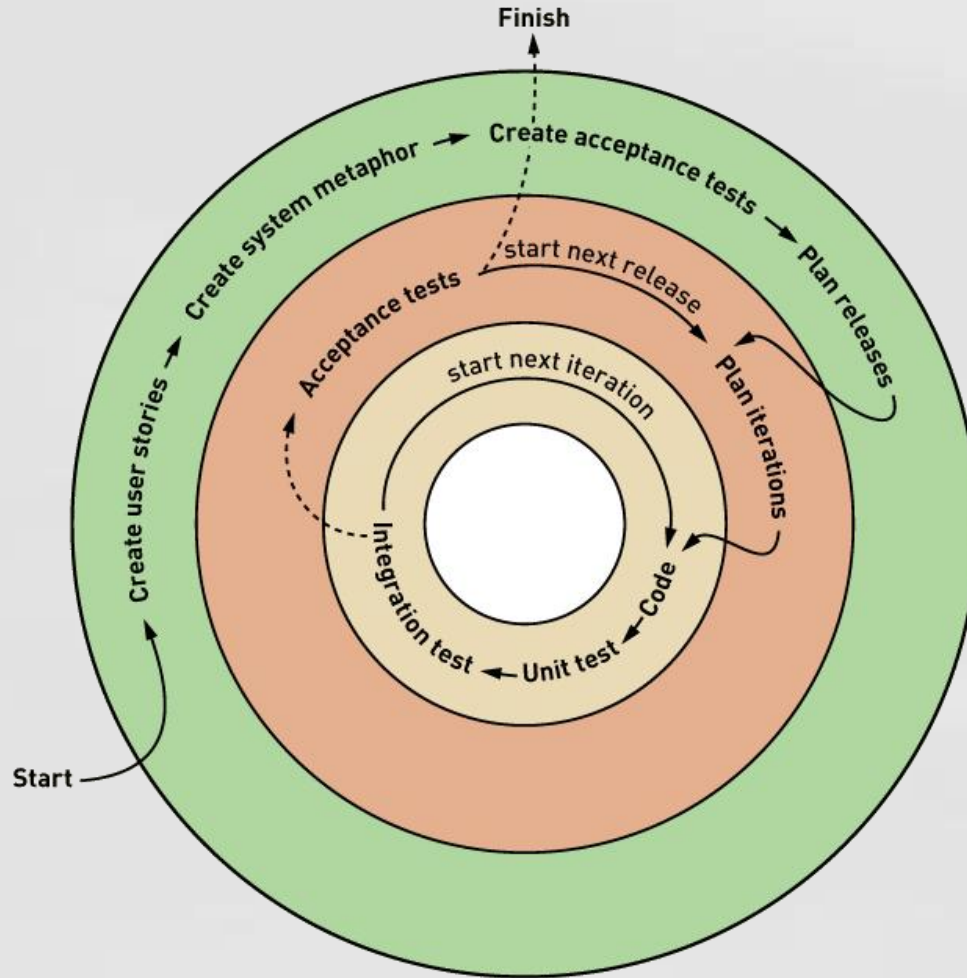
XP Core Values and Practices



XP Project Approach

- Three level approach (three rings)
 - Outside ring – create user stories and define acceptance tests
 - Middle ring – conduct tests and do overall planning
 - Inside ring – iterations of coding and testing

XP Project Approach

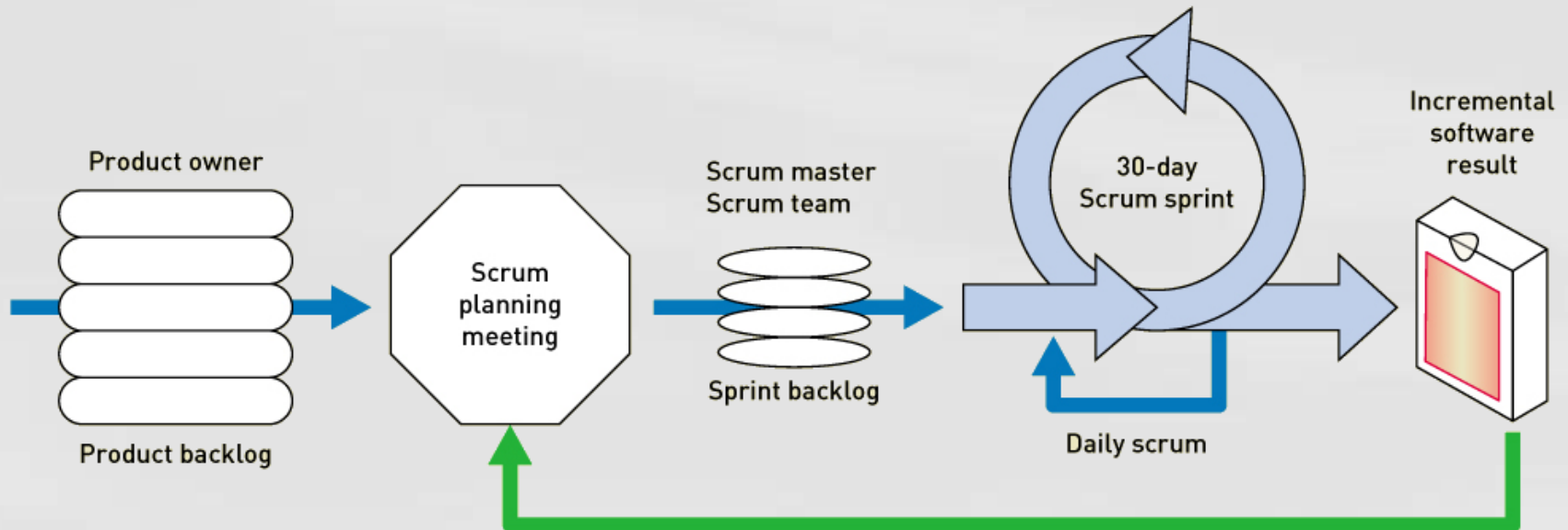


SCRUM

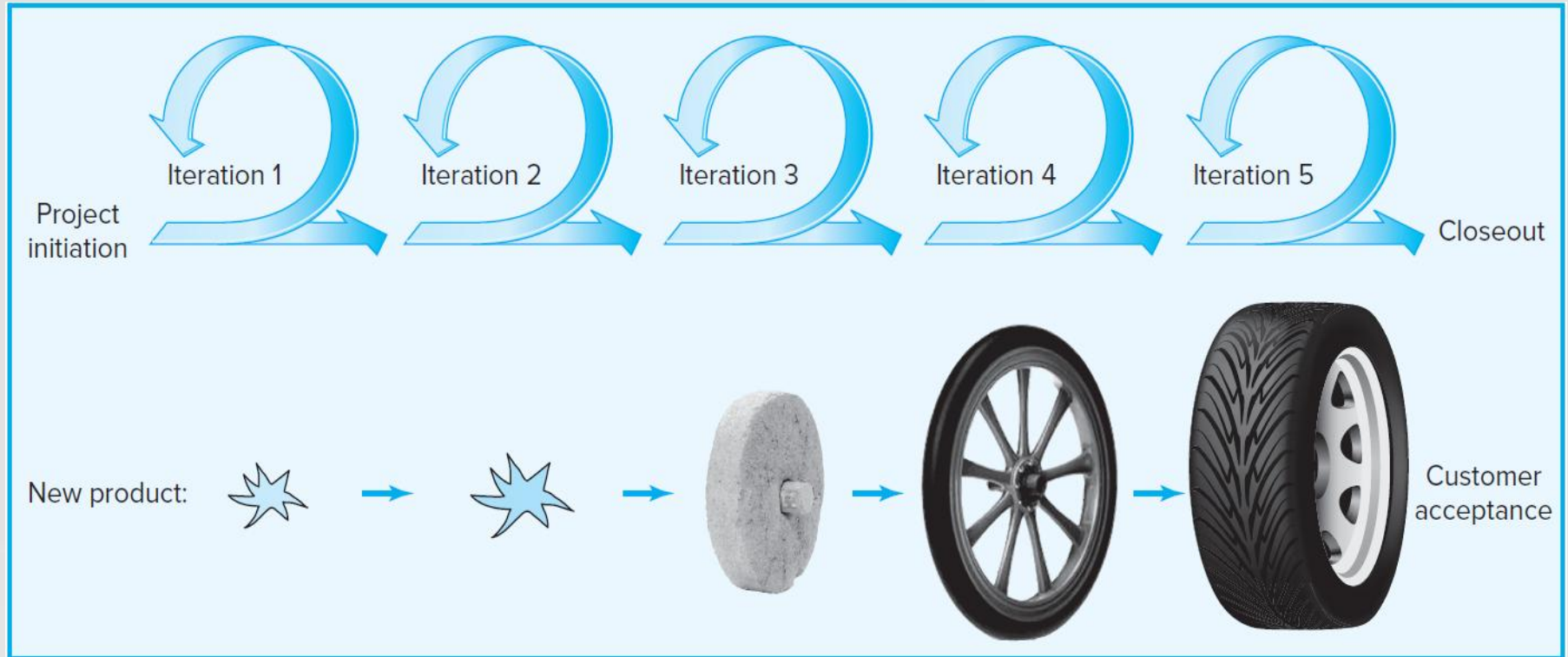
- An Agile framework for systems development
 - Intense effort involving the entire team for a defined period of time
- Product backlog
 - Prioritized list of user requirements
- Product owner
 - The client stakeholder who controls backlog
- Scrum master
 - Scrum project manager

Scrum Sprint

- Scrum sprint
 - A time-controlled mini-project to implement part of the system



Iterative, Incremental Product Development



Principles of Project Management:

The Need for Project Management

- Categories of project success
 - Successful projects – on time, within budget, on scope
 - Challenged projects – failure in one area
 - Failed projects – cancelled or not used
- Recent years have seen some improvement, but still 1/3 to 1/2 of projects are challenged or fail

The Need for Project Management

- Reasons for failure
 - Undefined project management practices
 - Poor IT management and poor IT procedures
 - Inadequate executive support for the project
 - Inexperienced project managers
 - Unclear business needs and project objectives
 - Inadequate user involvement
 - Projects not delivered on time, within budget, and scope

The Role of the Project Manager

- Project Management

- Organizing and directing other people to achieve a planned result within a predetermined schedule and budget
- The processes used to plan the project and then to monitor and control it.

- Project Manager

- Great need for effective project managers
- Internally managing people and resources
- Externally conducting public relations

Project Manager Responsibilities

● Internal Responsibilities

- Developing the project schedule
- Recruiting and training team members
- Assigning work to teams and team members
- Assessing project risks
- Monitoring and controlling project deliverables and milestones

● External Responsibilities

- Reporting the project's status and progress
- Working directly with the client (the project's sponsor) and other stakeholders
- Identifying resource needs and obtaining resources

Additional Project Stakeholders

- Client
 - the person or group that funds the project
- Oversight Committee
 - clients and key managers who review the progress and direct the project
- Users
 - the person or group of people who will use the new system

In-Class Task

Scenario

You are developing a new mobile app for a startup company that focuses on personalized fitness coaching. The project has a six-month timeline, with a changing feature set based on customer feedback.

- Each group should choose one methodology (UP, XP, or Scrum) for this project.
- Outline how the chosen methodology would be implemented across the project lifecycle. Include:
 - The approach to planning, designing, developing, testing, and delivering the product.
 - Specific practices that will be used (e.g., pair programming for XP, sprints for Scrum, phases in UP).
 - How client feedback will be incorporated throughout the project.

Presentation & Peer Review

- Each group will present their chosen methodology and plan.
- Other groups provide feedback, noting any potential challenges or alternative practices that could be considered.

Project Management Body of Knowledge (PMBOK)

It is a set of standard guidelines for project management.

PMPOK is organized into 10 knowledge areas:

- **Project Integration Management**—Integrating all the other knowledge areas into one seamless whole
- **Project Scope Management**—Defining and controlling the functions that are to be included in the system as well as the scope of the work to be done by the project team
- **Project Time Management**—Creating a detailed schedule of all project tasks and then monitoring the progress of the project against defined milestones
- **Project Cost Management**—Calculating the initial cost/benefit analysis and its later updates and monitoring expenditures as the project progresses
- **Project Quality Management**—Establishing a comprehensive plan for ensuring quality, which includes quality assurance and quality control activities for every phase of a project.

Project Management Body of Knowledge (PMBOK)

- **Project Human Resource Management**—Recruiting and hiring project team members; training, motivating, and team building; and implementing related activities to ensure a happy, productive team
- **Project Communications Management**—Identifying all stakeholders and the key communications to each; also establishing all communications mechanisms and schedules
- **Project Risk Management**—Identifying and reviewing throughout the project all potential risks for failure and developing plans to reduce or mitigate these risks
- **Project Procurement Management**—Developing requests for proposals, evaluating bids, writing contracts, and then monitoring vendor performance
- **Project Stakeholder Management**—Identifying and communicating with the stakeholders of the new system

Summary

- There are two approaches to the SDLC: Predictive and Adaptive
- A predictive SDLC, also known as the waterfall model, is used when it is possible to plan the project completely in advance
- An Adaptive SDLC, which uses iteration, is used when the requirements are less certain and the project will need to react to changes
- Agile development is the current trend in system development

Summary (continued)

- Unified Process is a formal iterative approach which uses UML models (with Agile philosophy) and UP disciplines
- Extreme Programming (XP) is an iterative approach which takes good practices to the extreme
- All System development projects use a methodology (or development process). A methodology includes an SDLC and tools, techniques, and models
- Project management is directing other people to achieve a planned result on schedule and on budget. Project managers have internal and external responsibilities.

Summary (continued)

- Project managers work with clients, who fund the project, an oversight committee which approves and reviews progress, and users who will directly interact with the system.
- The discipline of project management is organized into the Project Management Body of Knowledge (PMBOK) that includes ten knowledge areas. Project managers should study and digest this body of knowledge.
- Project management is an important and challenging career role. Information systems projects do not have a great success rate, and project management knowledge and skills are valued and needed to improve this record.

Questions?